

Do highly rated books tend to be more expensive?

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Abstract

Book prices may reflect popularity, demand, or perceived quality. Highly rated books could therefore be expected to have higher prices. Inspired by the work of Chevalier and Mayzlin (2004) on the influence of online reviews on book sales, this report investigates whether highly rated books also tend to be more expensive. The analysis relies on a dataset collected through web scraping from the website *Books to Scrape*. A Directed Acyclic Graph (DAG), descriptive visualizations, a boxplot, a t-test. The results suggest that the relationship between ratings and prices is weak and statistically insignificant.

1. Introduction

Book ratings are often associated with readers' satisfaction and perceived quality. Popular or highly appreciated books may attract more buyers and could therefore be sold at higher prices. However, pricing strategies may also depend on factors such as genre, publisher reputation, or author popularity.

This project investigates the following question: **Are better-rated books associated with higher prices?**

This study was inspired by the article *The Effect of Word of Mouth on Sales: Online Book Reviews* by Chevalier and Mayzlin (2006), which examines how online reviews influence book sales on digital platforms. Their work suggests that ratings and reviews can significantly affect consumer behavior and market performance. Inspired by this research, the present project explores a related question: whether higher ratings may also be associated with higher book prices.

Prediction: higher-rated books are associated with higher prices.

2. Data

The dataset was collected through **web scraping** from the online website: <https://books.toscape.com>

The final dataset contains **101 books** and includes **two principal variables** (see fig.2)

- **Rating:** Ordinal variable ranging from 1 to 5 stars.
- **Price:** Quantitative variable measured in British pounds (£).

And possible confounding variables include (see Fig. 1) genre, production cost, book length, author reputation and notoriety, as well as publisher reputation and prestige.

Figure 1. Conceptual DAG of the Relationship Between Book Ratings and Prices

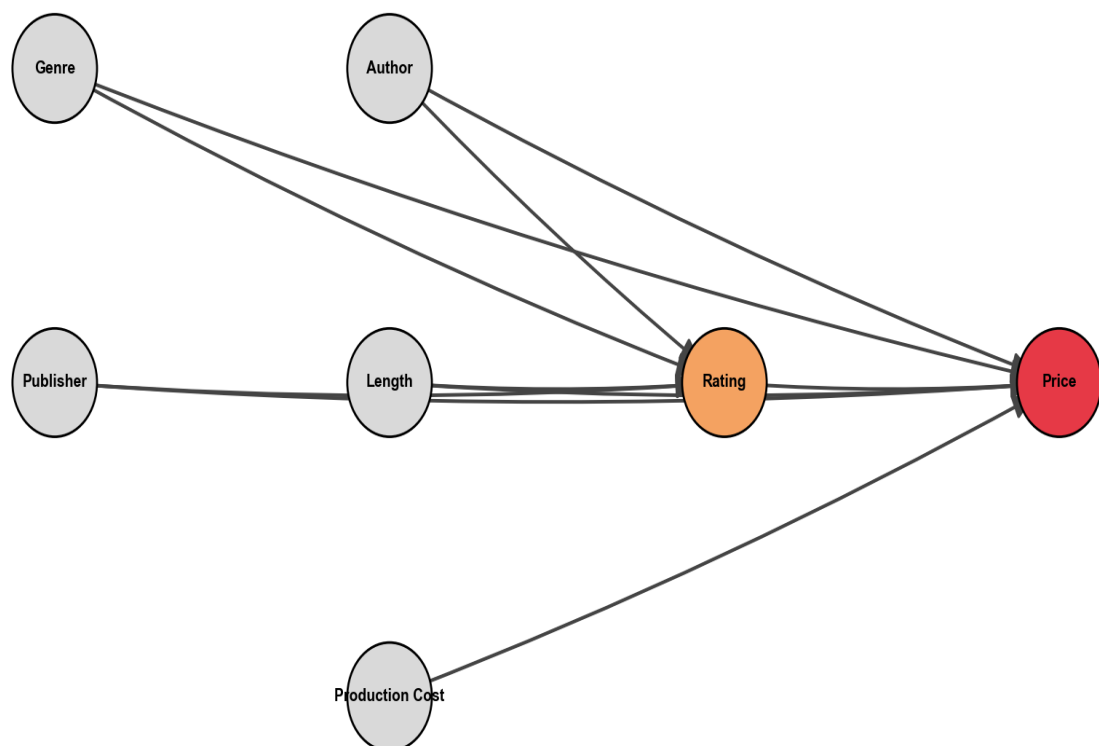
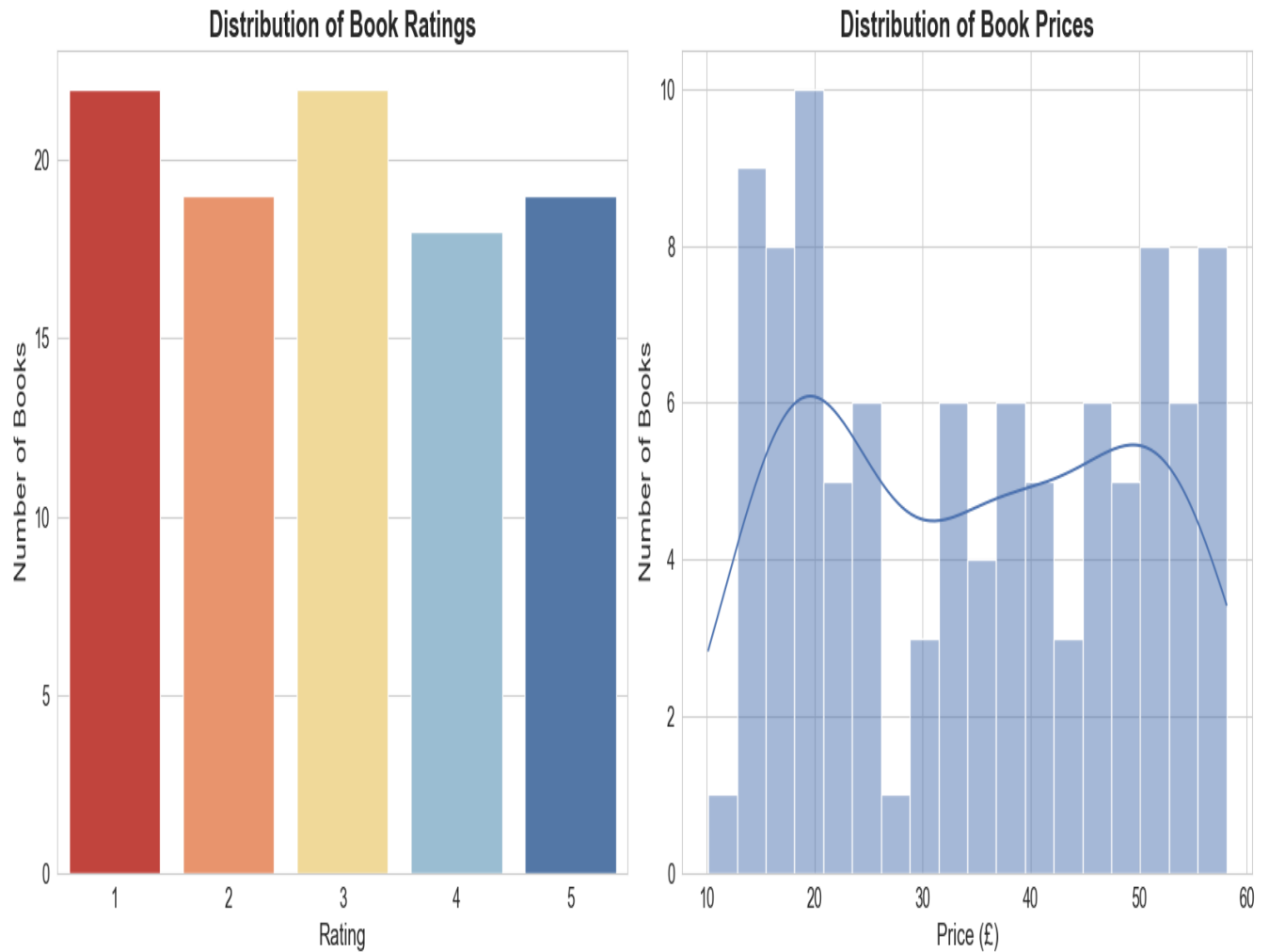


Figure 2. Descriptive Distribution of Ratings and Prices



3. Methods

To investigate the relationship between book ratings and prices, descriptive statistics and graphical visualizations were used. A **Directed Acyclic Graph**, **histograms**, and a **boxplot** were constructed to examine the distribution of ratings and prices. In addition, a t-test was performed to compare average prices across rating categories.

4. Results

The analyses suggest that the relationship between ratings and prices is weak. The boxplot does not reveal substantial differences in price distributions across rating categories.

Similarly, the t-test does not show statistically significant differences in average prices across rating groups ($p = 0.251$).

Consequently, the analyses **do not provide strong evidence** that highly rated books are more expensive (see Fig. 3).



5. Conclusion

While Chevalier and Mayzlin found that online reviews positively affect book sales, the present study does not find a statistically significant relationship between ratings and prices. Therefore, although highly rated books may attract more consumers and become bestsellers, they are not necessarily sold at higher prices.

This may be explained by the presence of additional confounding factors influencing book prices, such as genre, publisher reputation, author popularity, marketing strategies, production costs, edition type, or competitive pricing policies...

***Bibliography:** Chevalier, J. A., & Mayzlin, D. (2004). The Effect of Word of Mouth on Sales: Online Book Reviews. NBER Digest. National Bureau of Economic Research.*